Trainings ▾

Engine Noise Diagnostic Procedures - General Information

Note : When diagnosing engine noise problems, make sure that noises caused by accessories, such as the air compressor and power take-off, are not mistaken for engine noises. Remove the accessory drive belts to eliminate noise caused by these units. Noise will also travel to other metal parts not related to the problem. The use of a stethoscope can help locate an engine noise.

Engine noises heard at the crankshaft speed, engine rpm, are noises related to the crankshaft, rods, pistons, and piston pins. Noises heard at the camshaft speed, one-half of the engine rpm, are related to the valve train. A handheld digital tachometer can help determine if the noise is related to components operating at the crankshaft or camshaft speed.

Engine noise can sometimes be isolated by performing a cylinder cutout test. See Procedure 014-008 in the corresponding Troubleshooting and Repair Manual or Service manual for the engine being worked on. If the volume of the noise decreases or the noise disappears, it is related to that particular engine cylinder.

There is **not** a definite rule or test that will positively determine the source of a noise complaint.

Engine-driven components and accessories, such as gear-driven fan clutches, hydraulic pumps, belt-driven alternators, air-conditioning compressors, and turbochargers, can contribute to engine noise. Use the following information as a guide to diagnosing engine noise.

Main Bearing Noise

(Refer to Engine Noise Excessive - Main Bearing symptom tree)

The noise caused by a loose main bearing is a loud dull knock heard when the engine is pulling a load. If all main bearings are loose, a loud clatter will be heard. The knock is heard regularly every other revolution. The noise is the loudest when the engine is lugging or under heavy load. The knock is duller than a connecting rod noise. Low oil pressure can also accompany this condition.

If the bearing is **not** loose enough to produce a knock by itself, the bearing can knock if the oil is too thin, or if there is no oil at the bearing.

An irregular noise can indicate worn crankshaft thrust bearings.

An intermittent sharp knock indicates excessive crankshaft end clearance. Repeated clutch disengagements can cause a change in the noise.

Connecting Rod Bearing Noise

(Refer to Engine Noise Excessive - Connecting Rod symptom tree)

Connecting rods with excessive clearance knock at all engine speeds, and under both idle and load conditions. When the bearings begin to become loose, the noise can be confused with piston slap or loose piston pins. The noise increases in volume with engine speed. Low oil pressure can also accompany this condition.

Piston Noise

(Refer to Engine Noise Excessive - Piston symptom tree)

It is difficult to tell the difference between piston pin, connecting rod, and piston noise. A loose piston pin causes a loud double knock which is usually heard when the engine is idling. When the injector to this cylinder is cut out, a noticeable change will be heard in the sound of the knocking noise. However, on some engines the knock becomes more noticeable when the vehicle is operated on the road at steady speed condition.

Driveability - General Information

Driveability is a term that in general describes vehicle performance on the road. Driveability problems for an engine can be caused by several different factors. Some of the factors are engine-related and some are **not**.

Before troubleshooting, it is important to determine the exact complaint and whether the engine has a real driveability issue or if it simply does **not** meet driver expectations. The Driveability/Low-Power Customer Complaint Form is a valuable list of questions that **must** be used to assist the service technician in determining what type of driveability issue the vehicle is experiencing. Complete the checklist before troubleshooting the issue. The form can be found at the end of this section. If an engine is performing to factory specifications but does **not** meet the customer's expectations, explain to the customer that nothing is wrong with the vehicle and why.

Low power is a term that is used in the field to describe many different performance issues. However, in this manual low power is defined as the inability of the engine to produce the power necessary to move the vehicle at a speed that can be reasonably expected under the given conditions of load, grade, wind, and so on. Low power is usually caused by the lack of fuel flow that can be caused by any of the following factors:

- Lack of full travel of the accelerator pedal
- Failed boost sensor, if equipped
- Excessive fuel inlet, intake, exhaust, or drainline restriction
- Loose fuel pump suction lines.

Low power is the inability of the vehicle to accelerate satisfactorily from a stop or the bottom of a grade. Refer to the symptom tree Engine Power Output Low for the proper procedures to locate and correct a low-power issue. The chart starts off with basic items that can cause lower power.

Poor acceleration or response is described in this manual as the inability of the vehicle to accelerate satisfactorily from a stop or from the bottom of a grade. It can also be the lag in acceleration during an attempt to pass or overtake another vehicle at conditions less than rated speed and load. Poor acceleration or response is difficult to troubleshoot since it can be caused by factors such as:

- Engine- or pump-related factors
- Driver technique
- Improper gear shifting
- Improper engine application
- Worn clutch or clutch linkage.

Engine-related poor acceleration or response can be caused by several different factors such as:

- Failed boost sensor, if equipped
- Excessive drainline restriction
- Accelerator deadband.

Driveability/Low Power - Customer Complaint Form

Customer Name/Company ____Date ____________

- Describe Problem/Complaint____

- Symptoms of the Problem/Complaint
- When cranking:
- ____Cranks too slowly
- ____Cranks OK but does not start easily
- ____Cranks OK but does not start
- ____Slow start; _____ seconds
- ____Starts then dies
- ____Idle RPM is rough when engine is cold
- ____Idle RPM is rough when engine is hot
- When driving
- ____Misses or hesitates during acceleration
- ____Misses or hesitates during deceleration
- ____Stalls (dies) during acceleration
- ____Stalls (dies) during deceleration
- ____Smokes: ___ black ___ white
- ____Low power
- ____Unusual engine behavior

- When do you notice the Problem/Complaint occuring?
- Engine conditions:
- When the coolant temperature for the engine is:
- ___ cold ___ normal ___ hot ___ all temperatures
- When the engine is ____ RPM on the tachometer
- Weather conditions:
- ___ cold (below 10°C [50°F]) ___ hot (above 27°C [80°F]) ___ humid or rainy ___ other _______________
- When driving:
- ____Accelerating
- ____Decelerating
- ____Climbing a grade / hill
- ____Down hill
- ____Braking

- ____Unloaded
- ____Loaded

- How did the problem occur? Suddenly _______ Gradually _______
- At what hour/mileage did the problem begin? Hours _______ Miles_______ Since New _______
- After engine repair? Yes _______ No _______
- After equipment repair? Yes _______ No _______
- After change in equipment use? Yes _______ No _______
- After change in selected programmable parameters? Yes _______ No_______
- If so, what was repaired and when?____
- Does the vehicle also experience poor fuel economy? Yes _______No _______

Answer the following questions using selections (A through F) listed below. Circle the letter or letters that best describe the complaint.

A - Compared to fleet, B - compared to competition, C - compared to previous engine

D - Personal expectation, E - will **not** pull on hill, F - will **not** pull on flat terrain

- **A B C D E F**
- Can the vehicle obtain the expected road speed? Yes _______ No_______
- What is desired speed? rpm/mph _______
- What is achieved speed? rpm/mph _______
- Gross vehicle weight ______________
- **A B C D E F**
- Has the vehicle's load changed? Yes _______ No _______
- Is the vehicle able to pull the load? Yes _______ No _______
- When?
- _______ On hilly terrain
- _______ With a loaded trailer
- _______ On flat terrain
- _______ Other ____

IF THE ANSWER WAS NO TO ONE OF THE PREVIOUS QUESTIONS, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST AND GO TO THE LOW-POWER SYMPTOM TREE.

A B C D E F

- Is the vehicle slow to accelerate or respond? Yes _______ No _______

When?

- From a stop? Yes _______ No _______
- After a shift? Yes _______ No _______ rpm _______
- Before a shift? Yes _______ No _______ rpm _______
- No shift? Yes _______ No _______ rpm _______

- **A B C D E F**

- Does the vehicle hesitate after periods of long deceleration or coasting? Yes _______ No _______ rpm _______

IF THE ANSWER WAS YES TO ONE OF THE PREVIOUS QUESTIONS, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST, AND GO TO THE POOR ACCELERATION/RESPONSE SYMPTOM TREE.

Additional Comments:

- ____

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Driveability/Low Power/Excessive Fuel Consumption - Checklist

Vehicle/Equipment Specifications

Year, Type, and Model: ____

Transmission (RT 14609, and so forth): _____________________,

Duty Cycle:___________________________________,

Rear Axle Ratio, No. of Axles: _______, Application: Industrial ____, Marine ____, Genset ____, Automotive ____

Typical Gross Vehicle Weight: ______________________, Engine Rating:______________________

Trailer Type and Size: ___________________________________, Height:_________, Weight: _________

Tire Size (11R x 24.5, low profile, and so forth) ____

Tire Type: Radial _________, Standard Tread _________, Extra Tread_________

Fan Type: Direct Drive _________, Viscous _________, Clutch _________

Power Steering: Yes _______ No _______Air Conditioner: Yes _______No _______Air Shield: Yes _______ No _______Freon Compressor: Yes _______No _______

General Information					
DO Number:			SC Number:		
Fuel Pump Code:			Fuel Pump Serial Number:		
Mileage:			Engine Serial Number.:		
Date in Service:			Engine Model and Rating:		
Cruise Speed and rpm:			Rated Speed and rpm:		
Road Speed Governor:			Yes	No	Type:
Engine Brake:			Yes	No	Type/Brand:
Chassis and Other Related Items					
Tank Vents:	OK	**Not** OK	Obvious Fuel Leaks:	Yes	No
Brake Drag:	OK	**Not** OK	Axle Alignment:	OK	**Not** OK
Altitude:			Ambient Temperature:		
Fuel Heater:			Conditions (Wind, Rain, Snow):		
Fuel Type:			Number 1D	Number 2D	Other
Typical Terrain:		Flat	Hilly	Percent Asphalt	Percent Concrete

Additional Comments:

Note : Use this information for VE/VMS® run.

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Fuel Consumption - General Information

The cause of excessive fuel consumption is hard to diagnose and correct because of the potential number of factors involved. Actual fuel consumption problems can be caused by any of the following factors:

- Engine factors
- Vehicle factors and specifications
- Environmental factors
- Driver technique and operating practices
- Fuel system factors
- Low-power/driveability problems.

Before troubleshooting, it is important to determine the exact complaint. Is the complaint based on whether the problem is real or perceived, or does **not** meet driver expectations? The Fuel Consumption - Customer Complaint Form (on the next page) is a valuable list of questions that can be used to assist the service technician in determining the cause of the problem. Complete the form before troubleshooting the complaint. The following are some of the factors that **must** be considered when troubleshooting fuel consumption complaints.

1. **Result of a Low-Power/Driveability Problem:** An operator will change driving style to compensate for a low power/driveability problem. Some things the driver is likely to do are (a) shift to a higher engine rpm or (b) run on the droop curve in a lower gear instead of upshifting to drive at partial-throttle conditions. These changes in driving style will increase the amount of fuel used.
2. **Driver Technique and Operating Practices:** As a general rule, a 1-mph increase in road speed equals a 0.1 mpg increase in fuel consumption. For example, increasing road speed from 50 to 60 mph will result in a loss of fuel mileage of 1 mpg.
3. **Environmental and Seasonal Weather Changes:** As a general rule, there can be as much as a 1- to 1.5-mpg difference in fuel consumption depending on the season and the weather conditions.
4. **Excessive Idling Time:** Idling the engine can use from 0.5 to 1.5 gallons per hour depending on the engine idle speed.
5. **Truck Route and Terrain:** East/west routes experience almost continuous crosswinds and head winds. Less fuel can be used on north/south routes where parts of the trip are **not only** warmer, but also have less wind resistance.
6. **Vehicle Aerodynamics:** The largest single power requirement for a truck is the power needed to overcome air resistance. As a general rule, each 10-percent reduction in air resistance results in a 5-percent increase in mpg.
7. **Rolling Resistance:** Rolling resistance is the second largest consumer of power on a truck. The type of tire and tread design has a sizable effect on fuel economy and performance. Changing from a bias ply to low-profile radial tire can reduce rolling resistance by about 36 percent.
8. **Additional Devices Using the Same Fuel Source:** Additional devices may use the same fuel tank as the vehicle. For example, excessive use of generators or reefers can falsely indicate high fuel consumption.

Additional vehicle factors, vehicle specifications, and axle alignment can also affect fuel consumption. For additional information on troubleshooting fuel consumption complaints, see Troubleshooting Excessive Fuel Consumption, Bulletin 3387245.

Fuel Consumption - Customer Complaint Form

Customer Name/Company ____Date ______________

Answer the following questions. Some questions require making an X next to the appropriate answer.

1. What fuel mileage is expected? _______ Expected mpg
2. What are the expectations based on? Original mileage _______, Other units in fleet _______, Competitive engines _______Previous engine owned_______, Expectations **only** _______, VE/VMS® report _______
3. When did the problem occur? Since New _______, Suddenly _______, Gradually _______
4. Did the problem start after a repair? Yes _______ No _______If so, what was repaired and when? ____
5. Is the vehicle also experiencing a driveability issue (low power or poor acceleration/response)? Yes _______ No _______

IF ANSWERED YES, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST, AND GO TO THE ENGINE POWER OUTPUT LOW TROUBLESHOOTING SYMPTOM CHART.

1. Is the problem seasonal? Yes _______ No _______
2. Weather conditions during fuel consumption check? Rain _____, Snow_____, Wind _____, Hot temperatures _____, Cold temperatures _____
3. How is the fuel mileage measured? Tank _____, Trip _____, Month_____, Year _____Hubometer _____, Odometer _____
4. Are accurate records kept of fuel added on the road? Yes _______No _______
5. Do routes vary between compared vehicles? Yes _______ No _______
6. Have routes changed for the engine being checked? Yes _______ No_______
7. What are the loads hauled, compared to comparison unit? Gross Vehicle Weight _____________Heavier _______, Lighter _______
8. What is the altitude during operation? Below 10,000 feet _______, Above 10,000 feet _______
9. How much of the time is the truck spent idling? Hours/day _________
10. Is the driver technique or operating practices affecting fuel economy?

- High road speed: mph _________
- Operate at rated speed or above: rpm _________
- Incorrect shift rpm: Shift rpm _________, Torque peak _________
- Operate at a cruise speed: rpm _________
- Compensating for low power: Yes _______ No ______

IF, AFTER FILLING OUT THIS FORM, IT APPEARS THAT THE ISSUE IS NOT CAUSED BY VEHICLE FACTORS, ENVIRONMENTAL FACTORS, OR DRIVER TECHNIQUE, FILL OUT THE DRIVEABILITY/LOW-POWER/EXCESSIVE FUEL CONSUMPTION CHECKLIST, AND GO TO THE FUEL CONSUMPTION EXCESSIVE TROUBLESHOOTING SYMPTOM TREE.

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Oil Consumption

In addition to the information that follows, a service publication is available titled Technical Overview of Oil Consumption, Bulletin 3379214.

Cummins Inc. defines acceptable oil usage as outlined in the following table.

ACCEPTABLE OIL USAGE									
ANY TIME DURING COVERAGE PERIOD									
ENGINE FAMILY	MILES PER QUART	MILES PER LITER	MILES PER IMPERIAL QUART	KM PER QUART	KM PER LITER	KM PER IMPERIAL QUART	HRS PER QT	HRS PER LITER	HOURS PER IMPERIAL QUART
A - Series	-	-	-	-	-	-	10.0	10.6	12.0
B3.3/4B	400	425	475	650	675	775	10.0	10.6	12.0
ISF	400	425	475	650	675	775	10.0	10.6	12.0
6B/ISB/QSB	400	425	475	650	675	775	10.0	10.6	12.0
6C/ISC/QSC/ISL	400	425	475	650	675	775	10.0	10.6	12.0
V/VT-378	-	-	-	-	-	-	4.0	4.3	5.0
V/VT-504	250	265	310	400	425	485	4.0	4.3	4.8
V/VT-555	250	265	310	400	425	485	4.0	4.3	4.8

L10	500	530	620	800	850	970	7.0	7.4	8.4
M11/ISM	500	530	620	800	850	970	7.0	7.4	8.4
N14/NT	500	530	620	800	850	970	7.0	7.4	8.4
ISX/QSX/Signature™	500	530	620	800	850	970	7.0	7.4	8.4
V/VT/VTA-903	250	265	310	400	425	485	4.0	4.3	4.8
KT/KTA19	200	210	250	320	340	390	3.0	3.2	3.6
V/VT/VTA28	-	-	-	-	-	-	2.0	2.1	1.1
KT/KTA38	-	-	-	-	-	-	1.5	1.6	1.8
KTA50	-	-	-	-	-	-	1.1	1.2	1.3
QSK19	-	-	-	-	-	-	3.0	3.2	3.6
QST30	-	-	-	-	-	-	1.7	1.8	2.0
QSK23	-	-	-	-	-	-	1.7	1.8	2.0
QSK38	-	-	-	-	-	-	1.3	1.4	1.5
QSK45	-	-	-	-	-	-	1.25	1.3	1.5
QSK50	-	-	-	-	-	-	1.0	1.1	1.2
QSK60	-	-	-	-	-	-	0.9	0.95	1.1
QSK78	-	-	-	-	-	-	0.6	0.65	0.72

ACCEPTABLE OIL USAGE (Transit Bus, Shuttle Bus, and School Bus)

ANY TIME DURING COVERAGE PERIOD

ENGINE FAMILY	HRS PER QT	HRS PER LITER	HOURS PER IMPERIAL QUART	MILES PER QUART	MILES PER LITER	MILES PER IMPERIAL QUART	KM PER QUART	KM PER LITER	KM PER IMPERIAL QUART
B	10.0	10.6	12.0	200	210	240	320	340	385
C	8.0	8.5	10.0	150	160	180	240	255	290
L, M, N	4.0	4.3	5.0	100	105	120	160	170	195

Cummins®

Cummins
Engine Company, Inc.
Box 3005
Columbus, IN, U.S.A.
47202-3005

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<table>
<tr><td colspan="5">Engine Lubricating Oil Consumption Report</td></tr>
<tr><td rowspan="3">Owner's Name</td><td colspan="2">Engine Serial Number</td><td colspan="2">Engine Model and Horsepower</td></tr>
<tr><td colspan="4">Date of Delivery</td></tr>
<tr><td colspan="2">Month</td><td>Day</td><td>Year</td></tr>
<tr><td>Address</td><td colspan="4">Equipment Manufacturer</td></tr>
<tr><td>City</td><td>State/Province</td><td colspan="2">Equipment Serial Number</td><td>Fuel Pump Serial Number</td></tr>
<tr><td rowspan="2">Engine Application (describe)</td><td colspan="2">Oil and Filter Change Interval</td><td colspan="2">Complaint Originally Registered</td></tr>
<tr><td>Oil</td><td>Filters</td><td>Date</td><td>Miles/Hours/Kilometers</td></tr>
<tr><td colspan="5">Lubricating Oil Added</td></tr>
<tr><td colspan="2">Date Added Oil</td><td>Engine Operation Miles/Hours/Kilometers</td><td>Oil Added Liters/Quarts</td><td>Oil Used Brand/Viscosity</td></tr>
<tr><td colspan="2">Start Test</td><td></td><td></td><td></td></tr>
<tr><td colspan="2"></td><td></td><td></td><td></td></tr>
</table>

Engine Lubricating Oil Consumption Report

Last Mileage/Hours/Kilometers______________________ Minus Start Mileage/Hours/Kilometers______________________
Equals Test Mileage/Hours/Kilometers______________________ Divided by Oil Added______________________
Equals______________________
Usage Rate______________________

Customer Signature	Cummins® Dealer	Cummins® Distributor
Cummins Inc. Form 4755		

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OIL CONSUMPTION REPORT	
Customer Name:	D/r:
Engine Model:	Mi/Km/Hr:
Engine Serial Number:	CPL Number:
Vehicle Make/Model:	Date:
Signed: ____	